23 **A** Before the negative ions of mass m and charge q enter the metal container, its velocity v is found from the following:

$$qV = \frac{1}{2}mv^2 \qquad \text{or} \qquad v = \sqrt{\frac{2qV}{m}}$$

For the ions to travel undeflected,
electric force qE = magnetic force Bqv

$$q\,(\frac{V}{d}) = Bq\,(\sqrt{\frac{2qV}{m}}$$

$$\frac{V}{Bd} = \sqrt{\frac{2qV}{m}}$$

$$\frac{q}{m} = \frac{V}{2B^2d^2}$$

24 **B**